When a Jakarta Kilometer Isn't an Ambon Kilometer

Context-Aware Fleet Routing Across Indonesia's Asymmetric Geographies

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Abstract. Monolithic routing engines systematically underperform in Indonesia because the country's operating environments diverge beyond what a single parameterization can express: hyper-dense megacities such as Jakarta impose extreme node density, congestion-dominated travel times, and tight delivery windows, while archipelagic hubs such as Ambon impose sparse demand, topographic barriers, and schedule-locked maritime crossings. This paper introduces a *Context-Aware Parameterization Layer* for an open-source routing stack composed of OpenStreetMap (OSM) road data, the OSRM routing engine, the VROOM vehicle-routing optimizer, and Uber's H3 hexagonal spatial index. The framework classifies a target region into one of two archetypes and reconfigures every layer of the stack accordingly: H3 resolution, routing profile, and time-window semantics. We formalize the problem as a vehicle routing problem with time windows (VRPTW) augmented with canvassing coverage rewards, and we inject three physics-inspired mechanisms — a gravitational model of spatial demand for node aggregation, simulated annealing for global route search, and a topological-viscosity model of effective travel time that unifies urban congestion and maritime step delays in a single cost surface. We describe the execution pipeline, present the complete mathematical formulation, and propose an evaluation protocol contrasting the context-aware engine against a monolithic baseline on the Jakarta–Ambon archetype pair.

Keywords: vehicle routing, VRPTW, H3 spatial indexing, OpenStreetMap, VROOM, gravity model, simulated annealing, archipelagic logistics, Indonesia.

1 Introduction

Indonesia is the world's largest archipelagic state: more than 17,000 islands spanning three time zones, containing both one of the densest urban agglomerations on Earth (Greater Jakarta, over 30 million inhabitants) and some of the most sparsely connected territories in Southeast Asia (Maluku, Papua). For distribution businesses that combine *delivery* (serving confirmed orders at exact coordinates) with *canvassing* (dispatching sales representatives into promising territory to acquire new outlets), this geography poses a problem that is rarely acknowledged in the routing literature: the *same* fleet-optimization stack must operate across regions whose cost structures are not merely different in degree but different in kind.

A routing engine tuned for Jakarta treats distance as almost irrelevant and congestion as everything; an engine tuned for Ambon treats congestion as almost irrelevant and a missed ferry as catastrophic. A single, monolithic configuration — one spatial resolution, one routing profile, one time-window policy — necessarily fails at least one of these environments. We refer to this as the *regional divergence problem.*

This paper proposes a remedy that does not require abandoning the open-source stack that makes nationwide deployment economically viable. Instead of building separate engines per region, we insert a *Context-Aware Parameterization Layer* above the stack: a regional classifier that selects, per territory, (i) the H3 spatial-indexing resolution used to aggregate demand, (ii) the OSM/OSRM routing profile used to generate travel-time matrices, and (iii) the constraint semantics (soft vs. hard time windows) passed to the VROOM optimizer.

Our contributions are fourfold:

1. **A dual-archetype characterization** of Indonesian operating environments — the hyper-dense megacity and the archipelagic hub — with an analysis of the specific failure modes a monolithic engine exhibits in each (Section 3).
2. **A context-aware architecture** that reconfigures each layer of the H3/OSM/OSRM/VROOM stack per archetype (Section 4).
3. **A unified mathematical formulation** combining a VRPTW-with-coverage objective, regional complexity modifiers, and three physics-

inspired mechanisms: gravitational demand aggregation, simulated-annealing route search, and topological viscosity (Section 5).

4. **An evaluation protocol** with explicit hypotheses for testing the framework on the Jakarta–Ambon pair against a monolithic baseline (Section 8).

The remainder of the paper reviews the underlying technologies (Section 2), develops the archetypes and the architecture, presents the formulation, describes the execution procedure and field applications, and closes with limitations and future work.

2 Background and Related Work

2.1 Vehicle Routing with Time Windows

The vehicle routing problem (VRP) originates with Dantzig and Ramser [1] and generalizes the traveling salesman problem to fleets with capacity constraints. The time-windowed variant (VRPTW), for which Solomon [2] established the canonical benchmarks, additionally requires every visit to begin inside a customer-specific interval. VRPTW is NP-hard; practical solvers rely on construction heuristics followed by local-search metaheuristics [3, 4]. Our formulation extends VRPTW with an *optional-coverage* term that rewards visiting high-value canvassing territory, placing it in the family of team-orienteering and prize-collecting VRP variants.

2.2 The Open-Source Routing Stack

OpenStreetMap (OSM) [11] supplies the underlying road graph, including attributes critical to this work: `highway` class, surface quality, and — decisive for archipelagic regions — `route=ferry` relations with schedule metadata. **OSRM** [9] compiles OSM extracts into contraction hierarchies for millisecond-latency shortest-path and travel-time matrix queries; its Lua profile mechanism allows the same code base to produce radically different cost models (car-in-traffic vs. ferry-aware). **VROOM** [10] consumes those matrices and solves capacitated VRP/VRPTW instances with a state-of-the-art local-search pipeline. **Uber H3** [8] is a hierarchical hexagonal discrete global grid: at resolution 9 the average cell covers roughly $0.1\,\mathrm{km}^2$; at resolution 6, roughly $36\,\mathrm{km}^2$. Hexagons have uniform neighbor distance and near-isotropic adjacency, making them the natural aggregation unit for demand surfaces.

2.3 Gravity Models of Spatial Interaction

Gravity models — interaction proportional to the product of masses and inversely proportional to a power of separation — have a long lineage in retail and trade geography, from Reilly's law of retail gravitation [12] through Zipf's intercity-movement hypothesis [13] to Huff's probabilistic trading areas [14]. We repurpose this family as a *prioritization operator* over H3 cells, collapsing thousands of candidate micro-outlets into a tractable set of weighted centroids.

2.4 Physics-Inspired Metaheuristics

Simulated annealing (SA) [5, 6] transplants the Metropolis acceptance criterion [7] from statistical mechanics into combinatorial search: worse solutions are accepted with a probability that decays with a synthetic temperature, allowing early broad exploration and late greedy refinement. SA remains one of the most robust metaheuristics for asymmetric, discontinuous cost surfaces — precisely the surfaces produced by ferry schedules and rush-hour congestion.

3 The Dual-Archetype Problem

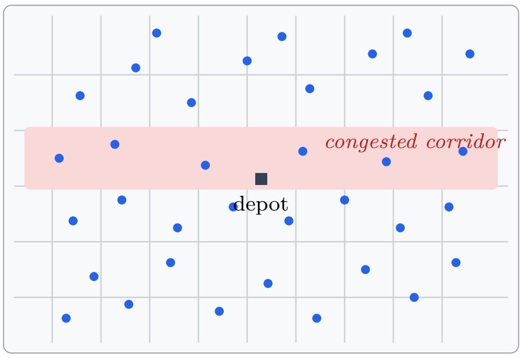
A monolithic routing engine fails in Indonesia because of *regional divergence*: the statistical and topological properties that drive routing cost differ so strongly between regions that any single configuration is badly mis-tuned for at least one of them. We anchor the analysis in two diametrically opposed archetypes, illustrated in Figure 1.

3.1 Archetype A: The Hyper-Dense Megacity (Jakarta)

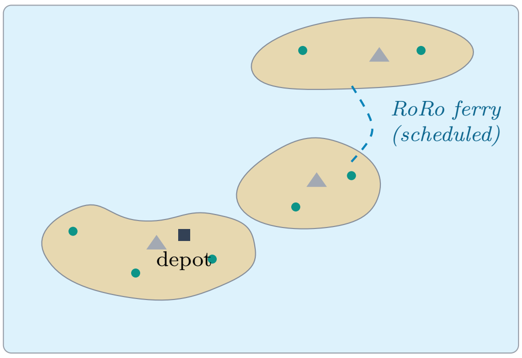
Greater Jakarta concentrates tens of thousands of serviceable outlets (*warung*, minimarkets, modern trade) into a contiguous conurbation. Its defining properties are: (i) **extreme node density** — hundreds of candidate visits per square kilometer; (ii) **congestion-dominated travel time** — the same 5 km leg can take 12 or 55 minutes depending on departure time; and (iii) **restrictive time windows** — modern-trade receiving docks and traffic-restriction schemes impose tight, dense windows. Distance is a poor proxy for cost; *time-of-day* is the dominant variable.

3.2 Archetype B: The Archipelagic Hub (Ambon)

Ambon and its satellite islands (Seram, Saparua, Haruku) invert every one of these properties: (i) **low node density** — demand is scattered



(a) Archetype A — megacity (Jakarta)



(b) Archetype B — archipelagic hub (Ambon)

Figure 1: The two operating archetypes. (a) Jakarta: extreme outlet density on a grid-like network where the dominant cost is congestion (shaded corridor), not distance. (b) Ambon: sparse coastal demand, topographic barriers (mountains), and inter-island legs that exist only as scheduled ferry crossings.

across coastal villages (*negeri*) separated by kilometers of empty road; (ii) **linear, topography-constrained networks** — routes follow coastlines around mountainous interiors, so the road graph is nearly one-dimensional and detours are catastrophic; and (iii) **schedule-locked maritime dependencies** — inter-island legs exist only at ferry departure times, converting the continuous routing problem into one punctuated by hard temporal barriers.

3.3 Failure Modes of a Monolithic Engine

Applying a single configuration across both archetypes produces four concrete failure modes:

1. **Resolution mismatch.** A fine H3 grid (res 9) over Ambon yields thousands of empty cells and fragments sparse demand; a coarse grid (res 6) over Jakarta merges distinct micro-markets into single cells, destroying the density signal canvassing depends on.
2. **Profile mismatch.** A car-in-traffic OSRM profile has no notion of ferry links or grade resistance, so Ambon inter-island legs are either unroutable or assigned absurd travel times; conversely a topography profile ignores the rush-hour dynamics that dominate Jakarta.
3. **Constraint-semantics mismatch.** Soft time windows are correct in Jakarta, where a 10-minute tardiness costs goodwill but not the route; they are dangerously wrong in Ambon, where missing a ferry cut-off strands a vehicle overnight.
4. **Cost-surface mismatch.** Jakarta's cost surface is smooth but time-varying; Ambon's is piecewise-flat with step discontinuities. A solver tuned for one surface explores the other inefficiently.

Table 1 summarizes the per-archetype configuration the framework applies to resolve these mismatches.

4 System Architecture: The Context-Aware Parameterization Layer

The framework adds one architectural element to the standard H3/OSM/OSRM/VROOM stack: a *Regional Classifier* that runs before any matrix generation or optimization, shown in Figure 2.

4.1 Regional Classification

The classifier consumes three cheap, data-derivable signals:

- **Demand density** d: registered outlets (or historical orders) per km^2 over the service territory;
- **Network connectivity** c: ratio of the largest connected road-graph component to total graph size, computed on the OSM extract *excluding* ferry edges — archipelagic regions fragment sharply under this test;
- **Maritime dependency flag** m: whether any demand lies on a component reachable only via `route=ferry` edges.

A territory with high d, $c \approx 1$, and $m = 0$ is assigned Archetype A; low d with $c \ll 1$ or $m = 1$ yields Archetype B. Intermediate regions (e.g., secondary cities such as Surabaya or Makassar) interpolate the parameter set; the two archetypes

Table 1: Context-aware parameterization of the stack across the two archetypes.

Component	Archetype A: Jakarta (megacity)	Archetype B: Ambon (archipelagic hub)
Uber H3 resolution	High resolution (res 8–9): hexagons of $\sim$0.1–0.7 km^2 capture granular store density and neighborhood-level canvassing.	Low resolution (res 6–7): hexagons of $\sim$4–36 km^2 match demand scattered thinly across coastal stretches and distinct villages (*negeri*).
OSM/OSRM routing profile	Traffic-heavy profile: time-dependent matrices simulating morning/evening rush hours and motorbike lane-splitting behavior.	Topographic/ferry profile: slope and surface penalties for steep terrain; `route=ferry` links admitted as valid, schedule-bound transit edges.
VROOM time windows	Soft constraints with buffers: high tardiness penalty, focus on dense, rapid sequencing of many small windows.	Hard constraints tied to cut-offs: daylight delivery hours, store closing times, and immutable ferry departure schedules.

studied here are the calibration anchors at the extremes.

4.2 Per-Layer Adaptation

H3 resolution. The classifier sets the aggregation resolution so that the *expected demand mass per cell* stays within a target band — fine cells in Jakarta preserve micro-market structure; coarse cells in Ambon avoid fragmenting village-scale demand (Figure 3).

OSRM profile. Archetype A compiles a traffic-heavy Lua profile with time-sliced speed classes per `highway` tag and a motorbike variant that discounts congestion (lane-splitting). Archetype B compiles a topographic profile that penalizes grade and unpaved surfaces and admits ferry edges with schedule-dependent effective costs.

VROOM constraint semantics. Archetype A submits jobs with soft time windows (violation penalties in the objective); Archetype B submits hard windows back-computed from immutable cut-offs — daylight hours, store closing times, and ferry departures.

5 Mathematical Formulation

We now formalize the optimization problem and the three physics-inspired mechanisms. Table 2 collects the notation.

5.1 The Global Objective Function

In a hybrid delivery–canvassing network the goal is not merely minimizing distance: it is *maximizing the value of territory covered* while respecting rigid time and capacity constraints. We minimize the total cost

Table 2: Notation.

Symbol	Meaning
V	set of exact delivery nodes (incl. depot)
H	set of H3 canvassing hexagons
K	set of vehicles / sales representatives
x_{ijk}	1 if vehicle k traverses arc (i, j)
y_h	1 if hexagon h is canvassed
C_{ijk}	contextual travel cost of arc (i, j) for k
ρ_h	opportunity cost of skipping hexagon h
W_h	demand mass of hexagon h
Ω	regional complexity factor
$T_{ij}, \widehat{T}_{ij}$	raw / adapted travel time
$\mu(t)$	time-of-day congestion coefficient
δ_{mar}	maritime step delay (wait + processing)
A_{ij}	gravitational attraction toward hexagon j
M_i, M_j	vehicle momentum / hexagon mass
ν	topological viscosity of the network
τ_{disc}	discrete (step) time penalty
$T^{(n)}$	annealing temperature at iteration n

$$Z = \min\left(\sum_{i\in V}\sum_{j\in V}\sum_{k\in K} C_{ijk}\, x_{ijk} + \sum_{h\in H} \rho_h\,(1-y_h)\right), \tag{1}$$

subject to the standard VRPTW structure. Every mandatory delivery node is served exactly once,

$$\sum_{k\in K}\sum_{i\in V} x_{ijk} = 1 \qquad \forall j \in V \setminus \{0\}, \tag{2}$$

flow is conserved per vehicle,

$$\sum_{i\in V} x_{ijk} = \sum_{i\in V} x_{jik} \qquad \forall j \in V,\ \forall k \in K, \tag{3}$$

vehicle capacity Q_k is respected,

$$\sum_{i\in V}\sum_{j\in V} q_j\, x_{ijk} \le Q_k \qquad \forall k \in K, \tag{4}$$

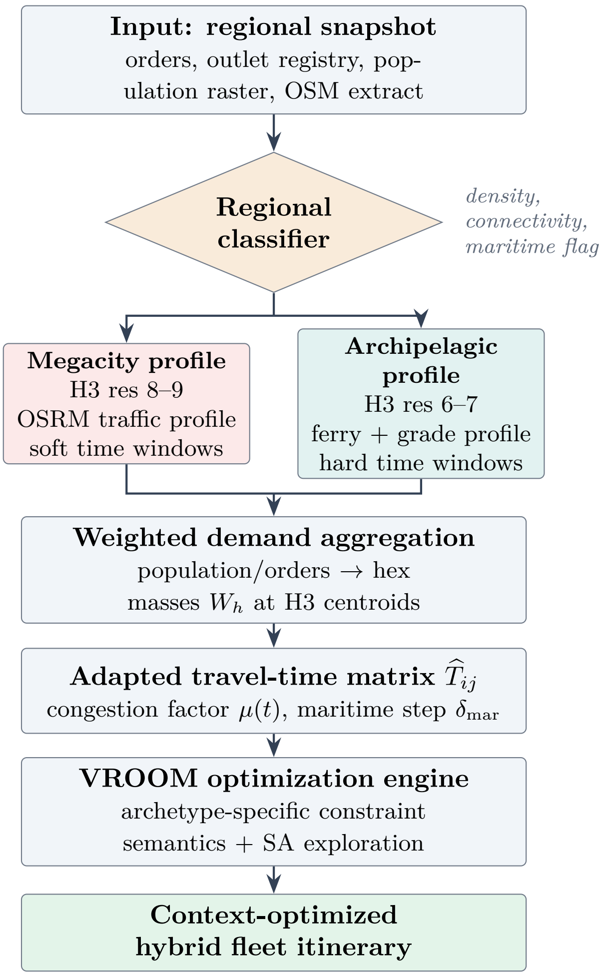


Figure 2: Context-aware execution pipeline. The regional classifier selects an archetype profile that reconfigures every downstream layer: spatial resolution, routing profile, and constraint semantics.

and visit times propagate along the adapted travel-time matrix,

$$t_{jk} \;\geq\; t_{ik} + s_i + \widehat{T}_{ij} - M\,(1 - x_{ijk}), \tag{5}$$

where s_i is service duration and M a large constant. Time windows $[a_j, b_j]$ are enforced with archetype-dependent semantics:

$$\begin{cases} \text{soft (A):} \quad Z \mathrel{+}= \sum_{j,k} w_j \max(0,\, t_{jk} - b_j), \\ \text{hard (B):} \quad a_j \leq t_{jk} \leq b_j. \end{cases} \tag{6}$$

A canvassing hexagon counts as covered only if some vehicle actually enters its centroid node $c(h)$:

$$y_h \;\leq\; \sum_{k \in K} \sum_{i \in V \cup c(H)} x_{i\,c(h)\,k} \qquad \forall h \in H. \tag{7}$$

Equation (1) creates a deliberate *tension system*: the solver wants to minimize travel cost C_{ijk} but is penalized ρ_h for ignoring high-value canvassing territory. The opportunity cost scales with the hexagon's demand mass, $\rho_h = \eta\, W_h$, with η a business-set conversion rate from demand mass to expected margin.

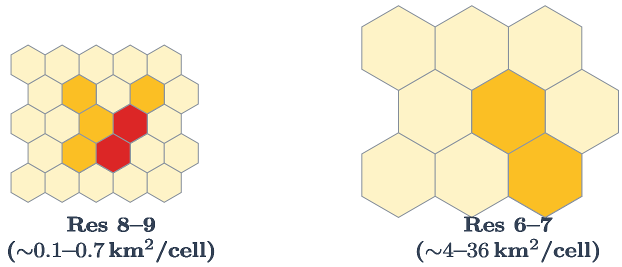


Figure 3: Archetype-dependent H3 resolution. Left: fine hexagons preserve Jakarta's micro-market density gradients (darker = higher demand mass W_h). Right: coarse hexagons match Ambon's village-scale demand without fragmenting it.

5.2 Regional Complexity Modifiers

The framework refuses to treat an Ambon kilometer like a Jakarta kilometer. Two modifiers implement this.

Demand mass. Each hexagon's mass composites population P_h, historical order volume O_h, and store density S_h, scaled by a regional complexity factor Ω_{region}:

$$W_h \;=\; \Omega_{\text{region}} \cdot \left[\lambda_P P_h + \lambda_O O_h + \lambda_S S_h\right], \tag{8}$$

where $\lambda_P, \lambda_O, \lambda_S$ are calibration weights. Ω_{region} captures the differential cost of serving a unit of demand in each region — effort per rupiah of opportunity is simply higher in archipelagic territory.

Adapted travel time. The raw OSRM matrix T_{ij} is transformed into the matrix actually given to the optimizer:

$$\widehat{T}_{ij} \;=\; T_{ij} \cdot (1 + \mu(t)) \;+\; \delta_{\text{mar}}(i,j), \tag{9}$$

with the congestion coefficient modeled as a baseline plus rush-hour peaks,

$$\mu(t) = \mu_0 + A_m e^{-\frac{(t-t_m)^2}{2\sigma_m^2}} + A_e e^{-\frac{(t-t_e)^2}{2\sigma_e^2}}, \tag{10}$$

where $t_m \approx$ 07:00–09:00 and $t_e \approx$ 16:00–19:00 are the morning and evening peaks. In Jakarta the amplitudes A_m, A_e are large enough that alternative local-road routing or motorbike dispatch becomes mandatory during peaks. In Ambon $\mu(t) \approx 0$, but δ_{mar} — fixed vessel wait plus port processing — acts as a heavy step function whenever a route crosses to Seram or Saparua.

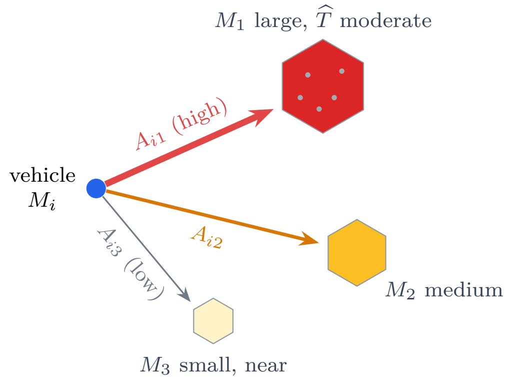


Figure 4: Gravitational prioritization over H3 centroids. Thousands of micro-outlets (small dots) are absorbed into hexagon masses M_j; the vehicle is pulled toward the highest mass-to-time ratio per Eq. (11), so only ∼50 centroid attractions are evaluated instead of $\sim 10^4$ point-to-point routes.

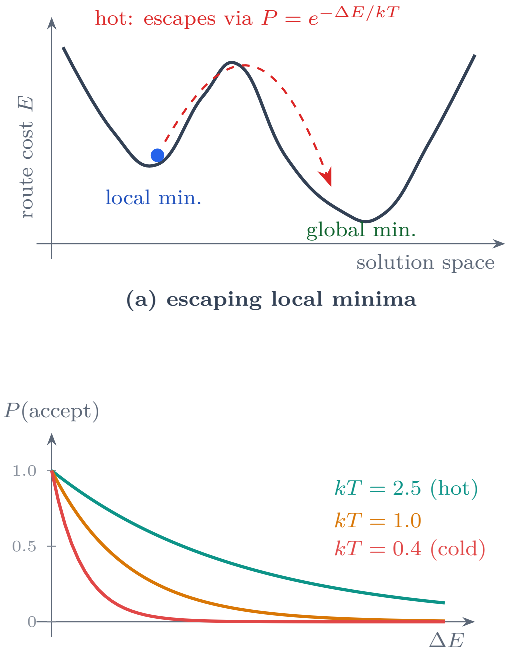
5.3 Gravitational Aggregation of Spatial Demand

Evaluating y_h against every individual store or prospective customer makes the coverage subproblem explode combinatorially — the VRPTW is already NP-hard, and canvassing multiplies the candidate set by orders of magnitude. We therefore collapse the demand surface into H3 centroids and rank them with a *gravitational interaction model*: each hexagon is a body of mass W_h (written M_j below), and the vehicle experiences an attraction

$$A_{ij} = G \frac{M_i^{\alpha} \cdot M_j^{\beta}}{\widehat{T}_{ij}^{\gamma}}, \tag{11}$$

where G is a strategic priority constant (raised during, e.g., new product launches), M_j the target cell's demand mass, M_i the vehicle's remaining capacity (its "momentum"), and α, β, γ elasticity parameters. Setting $\gamma > 1$ penalizes distant hexagons super-linearly and tightly clusters the resulting routes — the spatial analogue of preferring compact orbits.

Efficiency gain. Instead of computing routes to 10^4 individual micro-shops, the engine evaluates the pull of ∼50 H3 centroids (Figure 4); matrix generation cost falls from $O(|V_{\text{micro}}|^2)$ to $O((|V| + |H|)^2)$. The vehicle is mathematically drawn toward clusters with the highest mass-to-distance ratio, and individual outlets are resolved only *within* a hexagon the route has already committed to visit.



(a) escaping local minima

(b) Boltzmann acceptance, Eq. (12)

Figure 5: Simulated annealing over the routing landscape. (a) At high temperature the search escapes a locally optimal loop (e.g., South Jakarta) and discovers a superior basin (a loop including East Jakarta). (b) The probability of accepting a cost increase decays with ΔE and with cooling.

5.4 Simulated Annealing for Route Search

Even on the reduced node set, the visit-sequencing landscape over an asymmetric map is riddled with local minima — an acceptable loop in South Jakarta can mask a vastly superior loop through East Jakarta, and in Ambon a greedy sequence can strand the vehicle on the wrong side of a ferry cut-off. We therefore drive the improvement phase with simulated annealing. A temperature $T^{(n)}$ starts high and decays geometrically, $T^{(n)} = T^{(0)} c^n$ with $c \in (0.90, 0.99)$, and a candidate perturbation (2-opt, or-opt, cross-exchange) that changes the cost by ΔE is accepted with the Metropolis probability

$$P(\text{accept}) = \begin{cases} 1, & \Delta E < 0, \\ \exp\left(-\dfrac{\Delta E}{k\,T^{(n)}}\right), & \Delta E \geq 0, \end{cases} \tag{12}$$

with k a scaling constant. While hot, the search deliberately accepts worse routes and explores broadly across the map; as it cools it locks into the best basin found (Figure 5).

Efficiency gain. SA approximates near-optimal tours on both pathological surfaces — the discontinuous, ferry-punctuated landscape of Ambon and the time-varying congestion landscape of Jakarta — with a memory footprint linear in route length, which matters for nationwide batch optimization.

5.5 Topological Viscosity

Finally, we unify the two archetypes' cost structures in a single physical picture: the vehicle does not move through empty space but through a *medium with viscosity.* Effective travel time on an arc is

$$T_{ij}^{\text{eff}} = \frac{D_{ij}}{v_{\text{base}}} \cdot (1 + \nu_{\text{topology}}) + \tau_{\text{disc}}, \tag{13}$$

where D_{ij} is arc length, v_{base} the free-flow speed, ν_{topology} the *continuous* friction of the environment, and τ_{disc} a *discrete* step penalty. The two archetypes occupy opposite corners of this (ν, τ)-space:

- **Jakarta:** ν is high and time-varying (live traffic density is the dominant friction), while $\tau_{\text{disc}} = 0$ — there are no scheduled barriers, only continuous drag.
- **Ambon:** ν is low but nonzero (road quality, grade), while τ_{disc} is a massive step barrier — the RoRo ferry wait, which no amount of driving speed can amortize.

Figure 6 visualizes the resulting cumulative-time profiles. Note that Eq. (13) is the arc-level realization of the matrix transform in Eq. (9): ν implements $\mu(t)$ plus grade/surface friction, and τ_{disc} implements δ_{mar}. This ensures the optimizer never "draws lines between coordinates" — it evaluates the landscape as a physical environment with gravity, friction, and thermal state.

5.6 Synthesis: The Contextual Arc Cost

The arc cost consumed by the objective (1) composes the pieces above:

$$C_{ijk} = \phi_t \, T_{ij}^{\text{eff}}(k) + \phi_f \, F_k \, D_{ij}, \tag{14}$$

where F_k is the fuel/operating rate of vehicle class k (motorbike vs. light truck) and ϕ_t, ϕ_f convert time and fuel into a common monetary unit. The gravity scores A_{ij} do not enter C_{ijk} directly; they select *which* hexagon centroids are offered to the solver as optional high-reward jobs, keeping the instance compact.

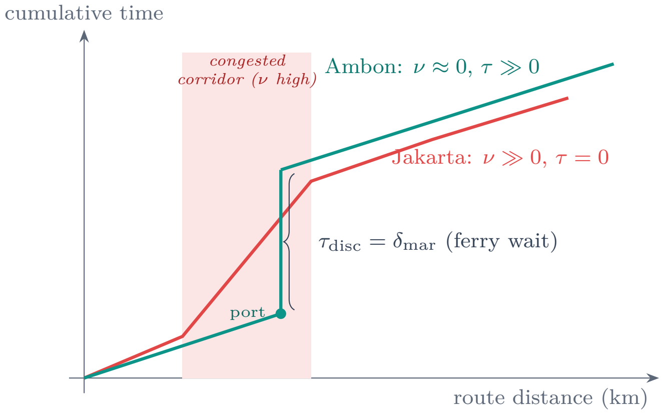


Figure 6: Topological viscosity, Eq. (13). Jakarta accumulates time through continuous friction concentrated in congested corridors; Ambon moves through a low-viscosity medium punctuated by a step barrier at the ferry crossing.

6 Context-Aware Execution Procedure

The full pipeline of Figure 2 executes as follows:

1. **Classify.** Compute density d, connectivity c, and maritime flag m from the outlet registry and OSM extract; assign the archetype and load its parameter profile (Table 1).
2. **Index.** Aggregate population, order history, and store registry into H3 cells at the profile's resolution; compute demand masses W_h via Eq. (8).
3. **Rank.** Score candidate canvassing cells with the gravity model, Eq. (11); retain the top-N centroids as optional jobs with rewards ρ_h.
4. **Route the graph.** Compile the archetype's OSRM profile; generate the raw matrix T_{ij} over delivery nodes plus retained centroids; apply Eq. (9) to obtain $\widehat{T}_{ij}$.
5. **Optimize.** Submit jobs, vehicles, capacities, and archetype-appropriate time windows (Eq. (6)) to VROOM; drive the improvement phase with the annealing schedule of Section 5.4.
6. **Guard.** For Archetype B, back-propagate latest feasible departure times from every ferry cut-off and daylight limit along the tentative route; if any guard is violated, re-optimize with the offending crossing moved to the next scheduled departure or deferred to a multi-day loop.
7. **Dispatch.** Emit the context-optimized hybrid fleet itinerary — per-vehicle sequences, ETAs,

and canvassing targets.

7 Strategic Field Applications

Jakarta canvassing. The engine prioritizes high-frequency, low-payload motorbike sales visits, packing dozens of small time-window jobs into tight geographic clusters. The gravity exponent $\gamma > 1$ keeps orbits compact; the soft-window penalty of Eq. (6) lets the solver trade a few minutes of tardiness for substantially denser sequencing; and the time-sliced matrix steers routes off arterial corridors during the $\mu(t)$ peaks of Eq. (10).

Ambon canvassing. The engine plans larger, multi-day loops. Hard windows anchored to ferry departures and daylight hours — propagated backward through Eq. (5) — prevent a vehicle from being stranded on the far side of the Leitimur or Hitu peninsula or on Seram after the last crossing. The coarse H3 grid ensures each stop on a loop represents a village-scale demand mass worth the detour, and Ω_{region} raises ρ_h so that remote-but-strategic territory is not perpetually starved by the travel-cost term.

Fleet composition. Because C_{ijk} is vehicle-indexed, the same optimization run allocates motorbikes to dense urban micro-routes and light trucks to island loops — the hybrid fleet emerges from the cost structure rather than being hand-assigned.

8 Proposed Evaluation Protocol

The framework is designed for empirical validation on the archetype pair. We propose the following protocol.

Data. OSM extracts for DKI Jakarta and Maluku (Geofabrik); outlet registries and historical order volumes from distributor records; population rasters for P_h; published ferry schedules for the Ambon–Seram and Ambon–Saparua crossings.

Baseline. The identical stack with a single monolithic configuration — H3 res 8, default OSRM car profile, uniform soft time windows — applied to both regions.

Scenarios. Matched daily instances per region: delivery-only, canvassing-only, and hybrid; weekday vs. weekend traffic in Jakarta; single-day vs. multi-day loops in Ambon.

Hypotheses.

H1. In Jakarta, the context-aware engine improves on-time arrival rate and weighted coverage at equal fleet size, by exploiting time-sliced matrices and fine-grained hexagon masses.

H2. In Ambon, the context-aware engine reduces ferry cut-off violations to zero while the monolithic baseline either strands vehicles or defensively under-serves outer islands.

H3. Gravitational aggregation reduces matrix-generation and solve wall-time by at least an order of magnitude relative to routing all micro-outlets, at negligible loss in realized coverage value.

Table 3: Evaluation metrics.

Metric	Captures
Total route duration	efficiency
On-time arrival rate	window fidelity
Ferry cut-off violations	stranding risk (B)
Weighted coverage $\frac{\sum_h W_h y_h}{\sum_h W_h}$	canvassing value
Matrix + solve wall-time	tractability
Vehicle-km per served demand unit	cost intensity

9 Discussion and Limitations

Data freshness. OSM coverage in eastern Indonesia is thinner and less frequently edited than in Java; grade and surface tags are incomplete, which degrades the Archetype-B viscosity term. Field GPS traces can progressively correct the profile.

Schedule volatility. Ferry timetables in Maluku shift with weather and season; δ_{mar} must be treated as a distribution, not a constant, and the guard step of Section 6 should plan against a conservative quantile.

Calibration burden. The elasticities (α, β, γ), mass weights $(\lambda_P, \lambda_O, \lambda_S)$, and Ω_{region} require historical conversion data to fit; until then they are expert-set priors. Sensitivity analysis over these parameters belongs in the evaluation protocol.

Classifier granularity. The two-archetype dichotomy is a deliberate simplification. Secondary cities blend both regimes; the parameter layer supports interpolation, but the interpolation policy itself is future work.

Stochasticity. SA is nondeterministic; production deployments should fix seeds per planning run and report cost distributions, not single draws.

10 Conclusion

We presented VRPTW-TERRA, a heterogeneous hybrid routing framework that adapts an entirely open-source stack — OSM, OSRM, VROOM, and Uber H3 — to the extreme regional divergence of Indonesian logistics. Rather than building per-region engines, the framework inserts a Context-Aware Parameterization Layer that classifies a territory and reconfigures spatial resolution, routing profile, and constraint semantics in one motion. A unified formulation couples a VRPTW-with-coverage objective to three physics-inspired mechanisms: gravitational demand aggregation that collapses intractable canvassing sets into weighted hexagon centroids, simulated annealing that survives the discontinuous cost surfaces of archipelagic geography, and a topological-viscosity model that expresses Jakarta's congestion and Ambon's ferry barriers as two corners of the same (ν, τ) space. The result is an engine that treats the landscape as a physical environment — with gravity, friction, and thermal state — rather than as empty space between coordinates. Empirical validation on the Jakarta–Ambon pair, per the protocol of Section 8, is the immediate next step.

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